Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 12)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

0 1 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 1 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

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10

0 2 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

0 2 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

0 2 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

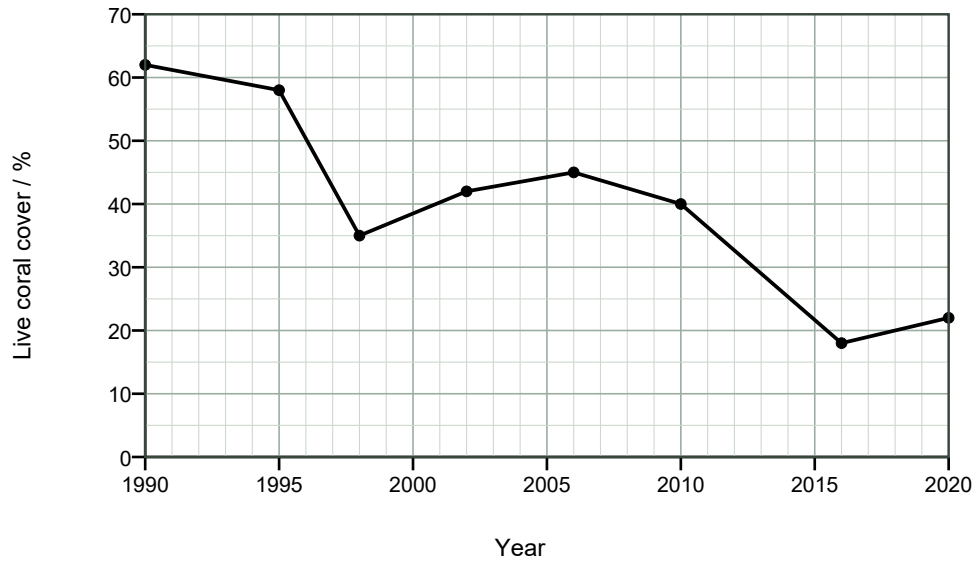
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10

0 3 Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 3 . 1 Describe the changes in coral cover shown in Figure 1.

[2 marks]

0 3 . 2 Explain what coral bleaching is and give **three** causes.

[4 marks]

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0 3 . 3 Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

0 3 . 4 Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

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0 4 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

0 4 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 4 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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0 5 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 5 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 5 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

0 5 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

2

0 5 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

10

0 6 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 6 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 6 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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5

0 7 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 7 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 7 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 8 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $m^3\ ha^{-1}\ yr^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

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0 8 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 8 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

0 8 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

10

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0 9 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 9 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 9 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

0 9 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 9 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

10

1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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10

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Write an essay on one of the following topics.

11 . 1 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

11 . 2 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS